Cluster Naming Results Report

Span2Type-LLM (MMR) — Ground-Truth vs. Predicted Names
CrossNER: All Five Domains

Span2Type Research

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Contents

1 Overview

This report presents the complete per-cluster naming results of **Span2Type-LLM (MMR)** across all five CrossNER target domains: AI, Literature, Music, Politics, and Science.

For each cluster the table shows:

- **Predicted Name** — the type label generated by Llama-3.1-8B (Ollama) from n=16 MMR-selected entity exemplars.
- **GT Label** — the ground-truth CrossNER entity type that contributes the most entities to the cluster (column-wise argmax of the confusion matrix).
- **Purity** — fraction of cluster entities belonging to the GT label.
- F_1 — BERTScore-F1 (RoBERTa-large) between predicted name and GT label.

Cell colour: green $F_1 \geq 0.99$ (near-perfect), yellow $0.90 \leq F_1 < 0.99$ (good), red $F_1 < 0.90$ (poor).

2 Summary

Table 1: Summary of Span2Type-LLM (MMR) cluster naming quality across all CrossNER domains (BERTScore-F1, RoBERTa-large). *Perfect* = $F_1 \geq 0.99$; *Low* = $F_1 < 0.90$.

Domain	#Clusters	Mean F_1	Perfect	Low
AI	14	0.9583	9	4
Literature	14	0.9787	11	2
Music	22	0.9391	11	9
Politics	24	0.9639	12	4
Science	22	0.9499	12	8
Overall	96	**0.9564**		

3 Per-Domain Results

Table 2: Ground-truth entity types vs. LLM-generated cluster names — AI domain (14 GT types, 12 unique generated names).

Ground-Truth Entity Types	**Generated Cluster Names (LLM)**
`algorithm`	*algorithm*
`conference`	*artificial intelligence*
`country`	*computer scientist*
`field`	*conference*
`location`	*country*
`metrics`	*field*
`misc`	*information retrieval*
`organisation`	*metric*
`person`	*organization*
`product`	*programming language*
`programlang`	*software*
`researcher`	*university*
`task`	
`university`	

3.1 AI Domain (14 clusters, mean $F_1 = 0.9583$)

Table 3: Span2Type-LLM (MMR) cluster naming results — AI domain. green $F_1 \geq 0.99$, yellow $0.90 \leq F_1 < 0.99$, red $F_1 < 0.90$.

C#	**Predicted Name**	**GT Label**	**Purity**	F_1
0	algorithms	algorithm	0.456	0.9999
1	fields	product	0.221	0.9994
2	organization	organisation	0.606	0.9994
3	computer scientist	researcher	0.699	0.8693
4	country	country	0.566	1.0000
5	programming language	product	0.305	0.8724
6	conference	misc	0.487	0.9994
7	information retrieval	task	0.672	0.8588
8	metrics	metrics	0.770	1.0000
9	university	location	0.474	0.9803
10	metric	metrics	0.735	0.9990
11	artificial intelligence	product	0.227	0.8390
12	software	product	0.560	0.9994
13	field	field	0.747	1.0000
Mean				**0.9583**

Table 4: Ground-truth entity types vs. LLM-generated cluster names — Literature domain (12 GT types, 11 unique generated names).

Ground-Truth Entity Types	Generated Cluster Names (LLM)
`award`	*academy*
`book`	*author*
`country`	*award*
`event`	*book*
`literarygenre`	*country*
`location`	*film festival*
`magazine`	*language*
`misc`	*literary work*
`organisation`	*mythology*
`person`	*novel*
`poem`	*theatre*
`writer`	

3.2 Literature Domain (14 clusters, mean $F_1 = 0.9787$)

Table 5: Span2Type-LLM (MMR) cluster naming results — Literature domain. green $F_1 \geq 0.99$, yellow $0.90 \leq F_1 < 0.99$, red $F_1 < 0.90$.

C#	Predicted Name	GT Label	Purity	F_1
0	book	book	0.295	1.0000
1	author	writer	0.754	0.9983
2	language	misc	0.825	0.9986
3	theatre	location	0.724	0.9789
4	academy	organisation	0.589	0.9982
5	mythology	book	0.364	0.9993
6	novel	book	0.308	0.9992
7	novels	book	0.610	0.9995
8	film festival	event	0.733	0.8766
9	country	country	0.515	1.0000
10	author	writer	0.760	0.9983
11	award	award	0.945	1.0000
12	novel	book	0.686	0.9992
13	literary work	literarygenre	0.748	0.8552
Mean				**0.9787**

Table 6: Ground-truth entity types vs. LLM-generated cluster names — Music domain (13 GT types, 14 unique generated names).

Ground-Truth Entity Types	**Generated Cluster Names (LLM)**
`album`	*album*
`award`	*award*
`band`	*band*
`country`	*country*
`event`	*ethnicity*
`location`	*genre*
`misc`	*music*
`musicalartist`	*music genre*
`musicalinstrument`	*musician*
`musicgenre`	*organization*
`organisation`	*record label*
`person`	*singer*
`song`	*song*
	theatre

3.3 Music Domain (22 clusters, mean $F_1 = 0.9391$)

Table 7: Span2Type-LLM (MMR) cluster naming results — Music domain. green $F_1 \geq 0.99$, yellow $0.90 \leq F_1 < 0.99$, red $F_1 < 0.90$.

C#	Predicted Name	GT Label	Purity	F_1
0	organization	organisation	0.821	0.9994
1	albums	album	0.701	0.9996
2	musician	musicalartist	0.787	0.8673
3	country	country	0.570	1.0000
4	music genres	musicgenre	0.773	0.8693
5	award	award	0.981	1.0000
6	musician	musicalartist	0.921	0.8673
7	music	musicgenre	0.905	0.8617
8	theatres	location	0.953	0.8413
9	record label	organisation	0.509	0.8457
10	band	band	0.702	1.0000
11	ethnicity	misc	0.627	0.9995
12	albums	album	0.588	0.9996
13	band	band	0.921	1.0000
14	band	band	0.721	1.0000
15	awards	award	0.427	0.9999
16	song	song	0.673	1.0000
17	singer	musicalartist	0.676	0.8708
18	genre	musicgenre	0.369	0.8534
19	music genre	album	0.286	0.8201
20	theatre	location	0.814	0.9789
21	album	song	0.268	0.9867
Mean				**0.9391**

Table 8: Ground-truth entity types vs. LLM-generated cluster names — Politics domain (9 GT types, 15 unique generated names).

Ground-Truth Entity Types	Generated Cluster Names (LLM)
`country`	*actor*
`election`	*ally*
`event`	*artist*
`location`	*city*
`misc`	*country*
`organisation`	*election*
`person`	*empire*
`politicalparty`	*historical figure*
`politician`	*nationality*
	organization
	party
	political party
	politician
	university
	war

3.4 Politics Domain (24 clusters, mean $F_1 = 0.9639$)

Table 9: Span2Type-LLM (MMR) cluster naming results — Politics domain. green $F_1 \geq 0.99$, yellow $0.90 \leq F_1 < 0.99$, red $F_1 < 0.90$.

C#	Predicted Name	GT Label	Purity	F_1
0	city	location	0.959	0.9799
1	artist	politicalparty	0.382	0.8670
2	election	election	0.990	1.0000
3	actor	person	0.937	0.9991
4	party	politicalparty	0.598	0.8785
5	nationality	misc	0.896	0.9995
6	city	location	0.988	0.9799
7	organization	organisation	0.908	0.9994
8	allies	event	0.444	0.9458
9	empire	country	0.878	0.9994
10	war	event	0.796	0.9469
11	political party	politicalparty	0.805	0.9304
12	historical figure	person	0.469	0.8508
13	university	organisation	0.774	0.9993
14	country	country	0.734	1.0000
15	party	politicalparty	0.953	0.8785
16	political parties	politicalparty	0.485	0.9185
17	politician	politician	0.880	1.0000
18	country	country	0.755	1.0000
19	politician	politician	0.704	1.0000
20	election	election	0.646	1.0000
21	city	location	0.941	0.9799
22	country	location	0.600	0.9802
23	politician	politician	0.735	1.0000
Mean				**0.9639**

Table 10: Ground-truth entity types vs. LLM-generated cluster names — Science domain (17 GT types, 19 unique generated names).

Ground-Truth Entity Types	Generated Cluster Names (LLM)
academicjournal	*actor*
astronomicalobject	*asteroid*
award	*astronomer*
chemicalcompound	*award*
chemicalelement	*biological process*
country	*building*
discipline	*continent*
enzyme	*element*
event	*field*
location	*historian*
misc	*journal*
organisation	*language*
person	*molecule*
protein	*planet*
scientist	*professional organization*
theory	*protein*
university	*scientist*
	thing
	university

3.5 Science Domain (22 clusters, mean $F_1 = 0.9499$)

Table 11: Span2Type-LLM (MMR) cluster naming results — Science domain. green $F_1 \geq 0.99$, yellow $0.90 \leq F_1 < 0.99$, red $F_1 < 0.90$.

C#	Predicted Name	GT Label	Purity	F_1
0	thing	misc	0.687	0.9991
1	astronomer	scientist	0.813	0.9970
2	protein	enzyme	0.410	0.9996
3	university	university	0.806	1.0000
4	elements	chemicalelement	0.848	0.8978
5	language	misc	0.708	0.9986
6	professional organization	organisation	0.862	0.8957
7	asteroid	astronomicalobject	0.890	0.8657
8	continents	location	0.756	0.9788
9	journal	misc	0.243	0.9991
10	field	discipline	0.473	0.9940
11	molecule	chemicalcompound	0.752	0.8385
12	planet	astronomicalobject	0.915	0.8666
13	biological process	misc	0.862	0.8804
14	scientist	scientist	0.722	1.0000
15	journal	academicjournal	0.841	0.8409
16	historian	scientist	0.868	0.9989
17	protein	misc	0.386	0.9993
18	actor	person	0.972	0.9991
19	building	location	0.692	0.9819
20	awards	award	0.956	0.9999
21	planet	astronomicalobject	0.942	0.8666
Mean				**0.9499**

4 Failure Analysis

Three recurring failure patterns account for most low-scoring clusters ($F_1 < 0.90$):

1. **Compound GT labels** (Music, Science): CrossNER uses concatenated labels such as `musicalartist`, `musicgenre`, `chemicalcompound`, `astronomicalobject`, and `academicjournal`. RoBERTa tokenises these differently from the plain-English predicted names (*musician*, *music genres*, *molecule*, *planet*, *journal*), reducing BERTScore even when the meaning is correct.

2. **Specificity mismatch** (AI, Literature, Science): The LLM generates a more specific or more general label than the GT label (e.g. *computer scientist* vs. `researcher`; *film festival* vs. `event`; *professional organization* vs. `organisation`).

3. **Mixed clusters** (AI, Politics): Low-purity clusters whose plurality GT label does not reflect the dominant entity concept (e.g. *conference* $\rightarrow$ `misc` in AI-C6; *artist* $\rightarrow$ `politicalparty` in Politics-C1).